EXACT NUMERICAL SOLUTION FOR THE DIFFUSED CHANNEL WAVEGUIDE

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We present an exact numerical solution based on a vector integral equation to investigate the diffused channel waveguide. Various examples are given and compared with other numerical and approximate methods. An approximate numerical solution is also given using the effective index method and the integral equation solution for the inhomogeneous slab.

1. Introduction

Channel waveguides obtained by diffusion are of considerable interest for the construction of components in integrated optics such as electrooptic modulators and switches [1,2].

Consequently the electromagnetic wave propagation in inhomogeneous bidimensional dielectric guides has evoked much interest in the recent years.

In 1969 Goell [3] presented his circular harmonic computer analysis of a rectangular homogeneous dielectric waveguide surrounded by a homogeneous medium, and Marcatili [4] presented his approximate analysis of rectangular homogeneous structure. After Knox and Toulios [5], Hocker and Burns [6] have used the effective index method with equivalent approximate method as W.K.B. to investigate the diffused channel waveguide. Hammer [7] has adopted the Marcatili's approach to obtain an approximate closed form solution for a gaussian index profile guide without sideways diffusion and Taylor [8] used a variational method with parabolic cylinder function.

A numerical method based on the finite element has been presented for a single mode waveguide by Yeh et al. [9,10], which can be used in a large variety of structures but the computation time and the memory storage can be very substantial near the cutoff.

In the last years numerical techniques have been used for a homogeneous or inhomogeneous optical waveguide of arbitrary shape surrounded by an uniform medium [11–14], and homogeneous guides in stratified media [15]. We present here a method based on a vector integral equation of the electric field to investigate inhomogeneous guides of arbitrary shape embedded in a substrate.

We have studied various types of guides and we have compared our results with other numerical techniques and approximate solutions.

In order to reduce the computation time which can be prohibitive for multimode guide, an approximate solution based on the effective index method with integral equation for the inhomogeneous slab guide [16] have been also developed.

2. Theory

The inhomogeneity considered here is shown on the fig. 1. It fills the $\mathcal{D}_2$ domain (medium II) with a permittivity profile $\epsilon_2(x, y)$ which can be a discontinuous or continuous function. It is surrounded by $\mathcal{D}_1$ (superstrate) and $\mathcal{D}_3$ (substrate) domains with constant permittivities respectively ϵ_1 and ϵ_3 ($\mu = \mu_0$ in each medium). $\mathcal{R}$ is the boundary between $\mathcal{D}_2$, $\mathcal{D}_1$ and $\mathcal{D}_3$ domains.

With a time dependence $e^{-i\omega t}$ and a propagation along z axis ($\boldsymbol{\mathcal{E}}(x, y, z) = \boldsymbol{E}(x, y)\, e^{i\beta z}$) the electric field $\boldsymbol{E}$ satisfies the Helmholtz equation in the distribution sense:

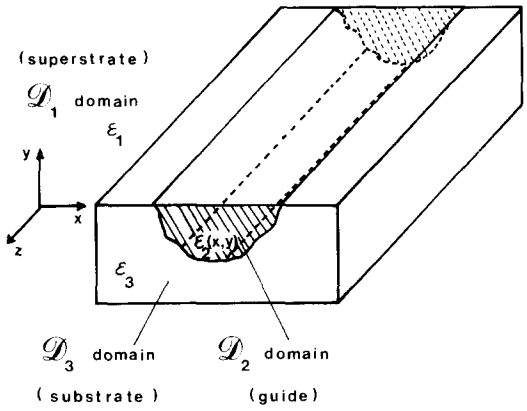

Fig. 1. Geometry of the problem: inhomogeneous dielectric waveguide embedded in a substrate.

$$(\Delta + (k_3^2 - \beta^2))E(x,y) = (k_3^2 - k^2(x,y)E(x,y) + \mathrm{e}^{-\mathrm{i}\beta z}\,\mathbf{grad}\left[\epsilon(x,y)\,\mathbf{grad}\left(\frac{1}{\epsilon(x,y)}\right)\cdot E(x,y)\,\mathrm{e}^{\mathrm{i}\beta z}\right], \tag{1}$$

with

$$k^2(x,y) = \begin{cases} k_1^2 & y>0\ ((x,y)\in\mathcal{D}_1), \\ k_2^2(x,y) & (x,y)\in\mathcal{D}_2, \\ k_3^2 & (x,y)\in\mathcal{D}_3, \end{cases} \tag{2}$$

$$k_j(x,y) = k_0 n_j(x,y),$$

$$n_j(x,y) = [\epsilon_j(x,y)/\epsilon_0]^{1/2}, \quad j = 1,2,3.$$

Using the vector Green's theorem, we have for the electric field $\boldsymbol{E}$, an exact integral equation:

$$E(x,y) = \iint_{\mathcal{D}_2} (k^2(x',y') - k_3^2)E(x',y') \times G(x,y;x',y')\,\mathrm{d}x'\,\mathrm{d}y' + \iint_{\mathcal{D}_2} \epsilon(x',y')\,\mathbf{grad}'\left(\frac{1}{\epsilon(x',y')}\right)\cdot E(x',y') \times \mathbf{grad}'\,G(x,y;x',y')\,\mathrm{d}x'\,\mathrm{d}y', \tag{3}$$

with in the distribution sense:

$$\epsilon(x,y)E(x,y)\cdot\mathbf{grad}(1/\epsilon(x,y)) = \epsilon(x,y)E(x,y)\cdot[\{\mathbf{grad}(1/\epsilon(x,y))\} + \boldsymbol{n}\sigma_{1/\epsilon}\delta_{\mathcal{R}}], \tag{4}$$

where $\{\mathbf{grad}(1/\epsilon(x,y))\}$ means that the derivation is to be taken in the function sense, $\sigma_{1/\epsilon}$ is the change of value of the function $1/\epsilon$ on the boundary $\mathcal{R}$. $\delta_{\mathcal{R}}$ is the cylindrical distribution defined on $\mathcal{R}$. The Green's function $G(x,y;x',y')$ is due to a line source embedded in the substrate (medium III) when the inhomogeneity is removed. It is given in terms of Fourier transform as:

$$y<0 \quad G(x,y;x',y') = \int_{-\infty}^{+\infty} \frac{\exp(-\gamma_1 y)}{\gamma_1+\gamma_3}\exp(\gamma_3 y')\exp[2\mathrm{i}\pi\nu(x-x')]\,\mathrm{d}\nu$$

$$y>0 \quad G(x,y;x',y') = \int_{-\infty}^{+\infty} \frac{1}{2\gamma_3}\left[\exp[-\gamma_3|y-y'|] + \frac{\gamma_3-\gamma_1}{\gamma_3+\gamma_1}\exp[-\gamma_3(y+y')]\right]\exp[2\mathrm{i}\pi\nu(x-x')]\,\pi\mathrm{d}\nu \tag{5}$$

with

$$\gamma_1^2 = 4\pi^2\nu^2 + \beta^2 - k_1^2 > 0,$$

$$\gamma_3^2 = 4\pi^2\nu^2 + \beta^2 - k_3^2 > 0. \tag{6}$$

In the medium III ($y>0$), $G(x,y;x',y')$ is expressed as the sum of two functions G_S and G_{NS}, with:

$$G_S(x,y;x'y') = \int_{-\infty}^{+\infty} \frac{1}{2\gamma_3}\exp(-\gamma_3|y-y'|)\exp[2\mathrm{i}\pi\nu(x-x')]\,\mathrm{d}\nu = (1/2\pi)K_0((\beta^2-k_3^2)^{1/2}[(x-x')^2+(y-y')^2]^{1/2}). \tag{7}$$

$K_0(x)$ is the modified Bessel function of zero order and first kind

$$G_{NS}(x,y;x',y') = \int_{-\infty}^{+\infty} \frac{1}{2\gamma_3} \frac{\gamma_3 - \gamma_1}{\gamma_3 + \gamma_1}$$

$$\times \exp[-\gamma_3(y+y')] \exp[2i\pi\nu(x-x')] \, d\nu .$$

The vectorial aspect of the integral equation (3) is due to the fact that the guided modes are hybrid. The waveguide supports the propagation of waves having two possible field configurations, classified as E^y_{pq} and E^x_{pq} modes [17] where the subscripts p and q indicate the number of extrema of the electric field in the x and y directions respectively.

To solve the vector integral equation (3), we used the classical moment methods [18] which lead to the linear system

$$AX = 0, \tag{8}$$

where X are the values of E_x and E_y expanded on suitable basis functions.

The values of β are the zeros of det (A) and then we can have the electric field distribution inside the guide with X. The size of the matrix A depends on the form of the basis function, the number of sampling points related to the guide transverse dimensions per wavelength, to the variation of the refractive index and to the studied mode structure. We have taken step functions as basis. In order to test the accuracy of the integral method, we have compared our results with those given by other numerical techniques: Yeh et al. (finite element method) [9,10], Goell (circular harmonic computer analysis) [3] and with the Marcatili's approximate method [4]. We have used an approximate method based on the effective index method [5,6] where the dispersion equation in case of diffused channel waveguide given by W.K.B. or equivalent equation [6] is replaced by a rigorous solution given by an integral equation [16]. We will call this approximate solution the "modified" effective index method.

3. Numerical results

Various examples are proposed here.

3.1. Homogeneous rectangular fiber

Yeh [9,10] using the finite element method, Goell [3] using harmonic expansion technique and Marcatili using approximate techniques obtained the dispersion characteristics of rectangular dielectric waveguides. We have compared those with the vector integral equation method. Results are shown in fig. 2 for the E^y_{11} mode. Excellent agreement was found with Goell's results and we note a difference between our results, Yeh and Marcatili near the cut-off.

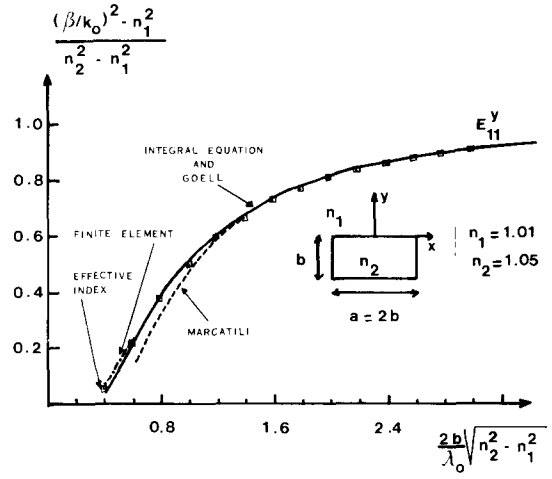

Fig. 2. Comparison of the vector integral equation results for the E^y_{11} mode with finite element method, Goell's results, Marcatili's approximate method and effective index method.

3.2. Homogeneous channel waveguide

We show in fig. 3 the comparison between our results and those given by Yeh and Marcatili and effective index method for the homogeneous channel waveguide with E^y_{11} mode. A good agreement is found with the vector integral equation and Yeh's results. At low frequencies Marcatili's curve deviate from Yeh's and our's results.

3.3. Inhomogeneous channel waveguides

3.3.1. Parabolic diffusion: E^x_{11} mode

We give here the dispersion curve for the E^x_{11} mode in the case of diffused channel waveguide with a unidirectional diffusion along the y-axis with a parabolic form of the refractive index $n_2(x,y) = n_3 + (n_{2m} - n_3)(b^2 - y^2)/b^2$ for $0 < y < b$, where b is diffusion depth, n_{2m} is the maximum refractive index at the surface and n_3 is the substrate refractive index. This

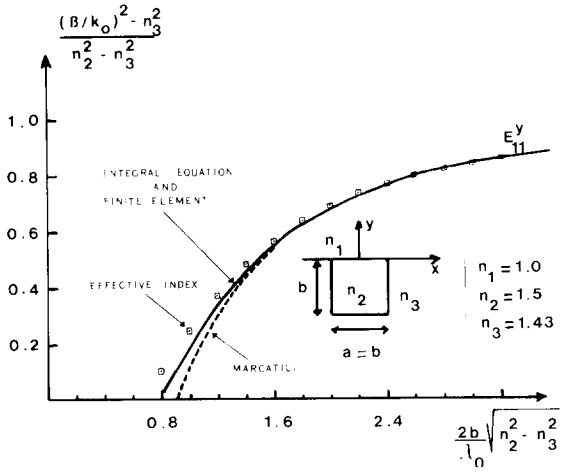

Fig. 3. Dispersion relation of an homogeneous channel waveguide. Comparison of the vector integral equation with finite element results, Marcatili's approach and the effective index method.

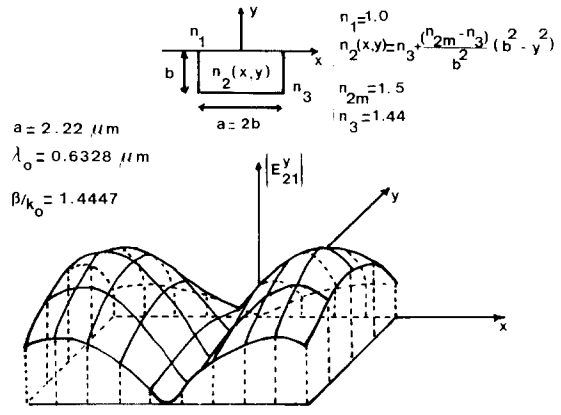

Fig. 5. Field distribution of the modulus of $E_y(x, y)$ for E_{21}^y mode at wavelength $\lambda_0 = 0.6328$ μm in a 1-D diffused channel waveguide with parabolic form of the refractive index. $a = 2b = 2.22$ μm.

index variation has been investigated by Yeh [10]. The diffusion process is assumed to be limited to the rectangular core region. The ratio of mask gap a to diffusion depth b is 2.

We have compared in fig. 4 the diffused channel waveguide (curve A) with an homogeneous channel

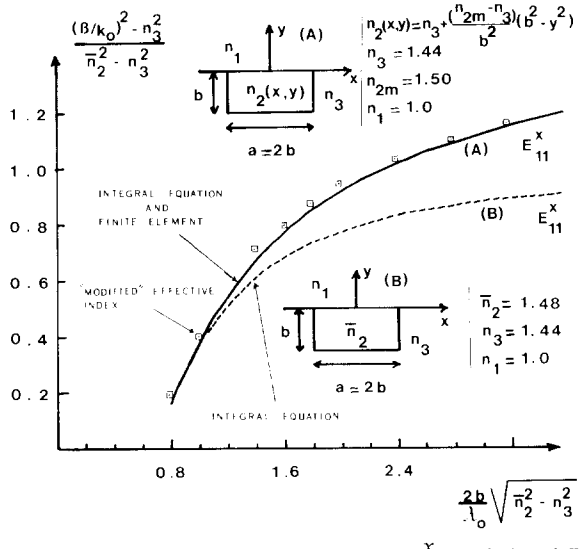

Fig. 4. Dispersion curves for the dominant E_{11}^x mode in a 1-D diffused channel waveguide and a uniform channel waveguide. Comparison of vector integral equation with finite element method and "modified" effective index method.

waveguide having the averaged index $\bar{n}_2$ of the diffused one inside the core ($\bar{n}_2 = 1.48$) (curve B). We have also compared with "modified" index method and Yeh's method. The agreement is good between Yeh's results and the vector integral equation. The "modified" index method gives a good approximation of the results. In fig. 5 we have represented the modulus $|E_y(x, y)|$ of the electric field for the E_{21}^y mode in the diffused channel waveguide with parabolic diffusion with a mesh of 50 points (10 on the x axis and 5 on the y axis) at 0.6328 μm (He-Ne laser wavelength) and with dimension $a = 2.22$ μm.

3.3.2. Circular diffusion: E_{11}^x mode

We examine here the case of a circular form for the refractive index $n_2(x, y)$. As before we have taken a rectangular core region surrounded by uniform substrate. The refractive index is described as $n_2(x, y) = n_3 + [(n_{2m} - n_3)/L^2]\,(L^2 - x^2 - y^2)$ where n_{2m} is the maximum index at the origin, n_3 the substrate index and $L^2 = b^2 + x^3$ if $y > x$ and $L^2 = (a/2)^2 + y^2$ if $y < x$, a and b are the width and depth of the core. L is the length of the straight line from the origin to the boundary. We have also compared with an homogeneous channel waveguide having the averaged index $\bar{n}_2$ of the diffused one ($\bar{n}_2 = 1.47$) (curve B), fig. 6.

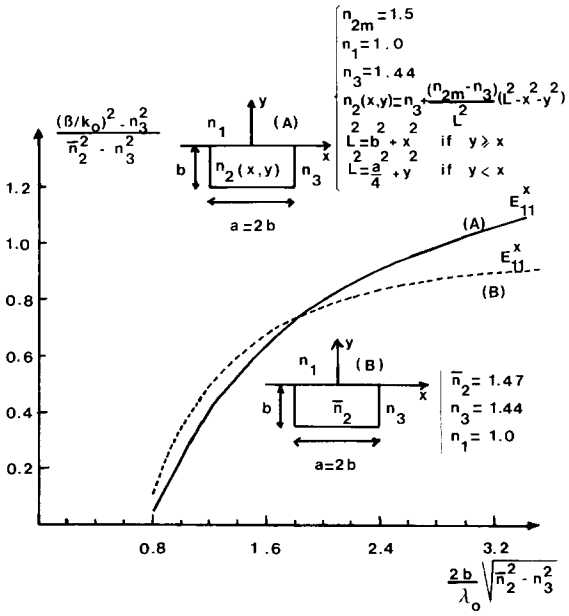

Fig. 6. Dispersion curves for the dominant E_{11}^x mode in a 2-D diffused channel waveguide and in a uniform channel waveguide.

4. Conclusions

We have developed a general and versatile analytical–numerical method using a vector integral equation to analyse the treatment of the wave propagation problem in an inhomogeneous guide embedded in a substrate. Not only are the propagation constants available by this method but we can have also the field distribution inside and outside the guide.

The accuracy of the obtained values can readily be controlled by increasing the number of the sampling points as long as round off errors are reasonable and limited by computer size machine. Thus the computation time is low for a guide supporting the dominant modes E_{11}^x and E_{11}^y (a few seconds for computing β/k_0 and field distribution on Amdahl 470/v7) but it can be prohibitive for multimode guides. Such an approximate method has been developed using the effective index method and the integral equation for an inhomogeneous slab waveguide which reduces the computing time and memory storage while giving good results.

The vector integral equation method can be extended to an inhomogeneous waveguide of arbitrary shape embedded in stratified media by changing the Green's function and toward microwave and submillimeter integrated circuits. It can be also used for the diffraction of two dimensional objects with an arbitrary three dimensional illumination angle.

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